Pair-46 factor-base campaign (2026-10-15)

Study: `research/pair46_factor_bases_20261015/`. **Branch:** `claude/pair-46-elliptic-curves-71ny4x`. Pair 46 has $p = 5t^2 + 19t + 19$ with $t = 34489701$, a 53-bit prime $N = p + 1 - t$, a CM(-19) crater with $j = -96^3$, and a floor curve. There were 14 items, each with a preregistered protocol, a run, a results report and an independent red-team review. Every verdict below is the red team's corrected verdict. 12 of the 13 items that ran were upheld, item 14 was refuted (`exact_obstruction` becomes `empirical_negative`), and item 11 was blocked.

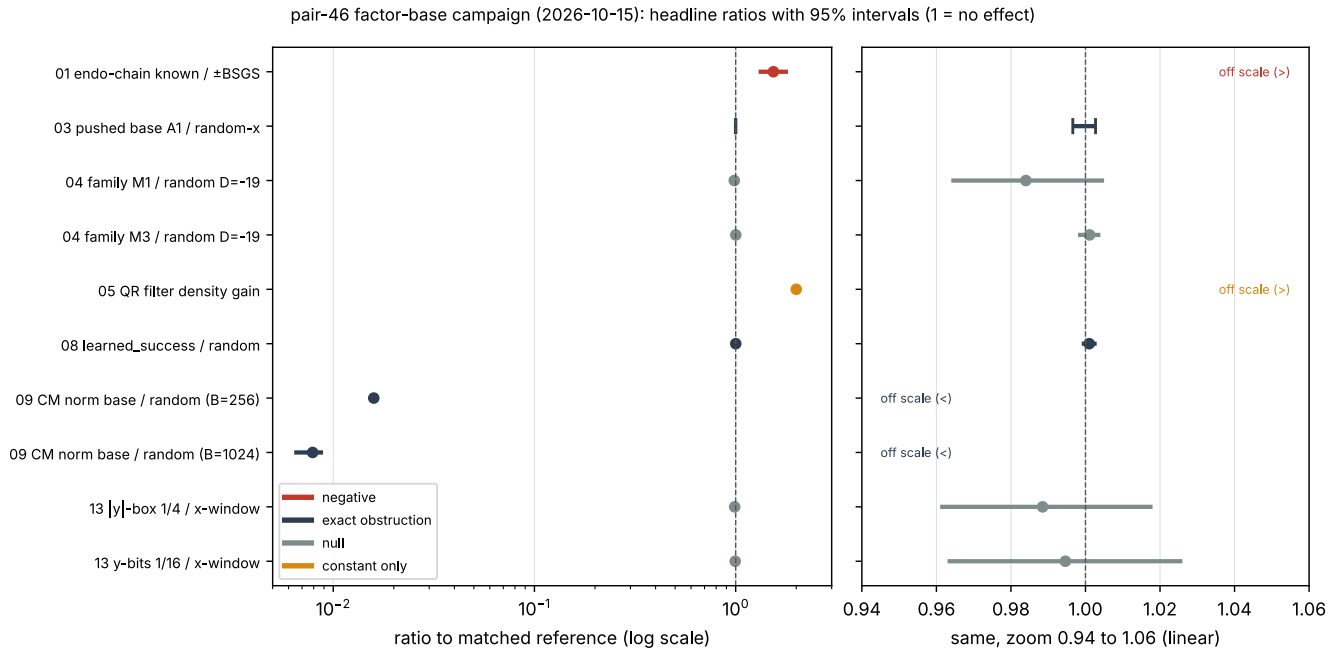


Figure 1: Headline ratios against each item's matched reference, with 95% intervals as recorded in the item's RESULTS.md (no recomputation). Item 01 is a cost ratio, so values above 1 are worse. The rest are yield, coverage or density ratios. Data: `summary_data.json`. Source: `plot_summary.py`.

Summary

#	item	verdict	headline (measured / exact)
01	endo-chain	negative	ψ-chain S = 1.47 [1.28, 1.69] against ±BSGS 0.955 [0.86, 1.05] (ratio 1.54 [1.30, 1.82]). Exact: a known-head chain is BSGS with $c_E \approx 1$.
02	deuring-lift	exact obstruction	The lift is 361a1 with rank 1, so lifting a target is the DLP. Twist bases give 0 Q-relations. λ1/GH difference +0.14 [−0.18, +0.26] against random.
03	floor-pullback	exact obstruction	φ is an isomorphism, so the yield is identical to the crater's. Toy ratio 0.9978 to 1.0023, |z| ≤ 1.40; the positive control had |z| ≥ 562.
04	family-sweep	null	M1 0.984 [0.964, 1.005], M3 1.0011 [0.998, 1.004], 20 to 30 bits. All positive controls had p ≤ 0.0005.
05	qr-window	constant only	Density gain 2.00004 [1.99967, 2.00042]. The relation ratio is 1 exactly (8000/8000).
06	pkm-rational-map	negative	Cost per system about $p^{2.0}$ (local toy slope), against $p^{0.52}$ for rho. About 4e6× rho at n1 = 5. Yield equals the flat set exactly.
07	torsion-division	exact obstruction	No ψ_m root on E. Preimage bases change coverage by exactly 0. Size/degree ρ ≤ 0.25.
08	learned	exact obstruction (counting); empirical part inconclusive	Learned selectors are within ±0.5% of random. $\sum_R c(R) = \|S\|^m$. The advantage is at most about μ/(1−e^(−μ)).
09	cm-norm	exact obstruction	$F_X + F_X \subseteq F_{4X}$. Coverage is 0.0159 [0.0156, 0.0161]× random at B = 256.
10	bounded-degree	exact obstruction	|F|/p ≥ 0.167 for d ≤ 8. |F| ≤ √p needs d ≥ √p/5.
11	ff-xedni	blocked	Not run. B2 and function-field xedni remain open.

#	item	verdict	headline (measured / exact)
12	model-small-x	exact obstruction	No Montgomery, Edwards or Hessian model over $\mathbb{F}_p$ for pair 46. Toy tests within ±1% of random.
13	xy-windows	null	Matched-size y-windows give D/D(X) 0.986 to 0.995, CIs within [0.956, 1.026], at 4 to 64× the scan price.
14	dor-verdict	**empirical negative** (refuted)	Binary toy model. The abandon rule fires for generic bases (0.86 to 1.36 per doubling) but never for the standard window. E1 fails for the degree-2 Macaulay rank on 937 of 4638 targets. Does not transfer to pair 46.

Conclusion, as strong as the evidence

- **No item found a factor-base choice that gives index calculus an advantage on pair 46 or on its toy analogues.**
- **The exact obstructions are scoped** to the tested families and metrics. They do not establish that no factor base can help. For example, bases chosen to make solving cheaper (item 08) and compact high-genus descriptions (item 10) are not covered.
- **The empirical findings are toy-scale** (12 to 30 bits, m = 2). The nulls have a resolution of about ±2 to 3%.
- **The cost results 01 and 06 are constants or local slopes**, not asymptotic exponents.
- **Not tested:** m ≥ 3, except through exact identities; relation collection with linear algebra at pair-46 size; F4/msolve; and item 11.

Positive-control status (item 05)

The positive control was detected. The predicted 2× density gain was measured at 2.00004 [1.99967, 2.00042], and the negative controls gave 1. The agreement of the main checks holds by construction. The independent validation is the E1 full-field counts and the A2/A3 halving controls, which keep about 1/4 of the m = 2 decompositions. Every other item that ran had its own positive control detected. The weak points are 08 (the random2 null failed at 16 bits) and 06 (the preregistered sugar-degree match failed; the amended lcm-degree match passed).

Blocked and cut

- **11, blocked:** no artifacts.
- **Cut:**
  - 03: no pair-46-size isogeny.
  - 04: no floor members, and no M1/M2 at 53 bits.
  - 06: chained arms stopped at n1 = 5, and there was no F4.
  - 07: no Gröbner invariants.
  - 14: m = 3, sym-S4 and n = 131 were not ported.
- **Hosts:** timings come from a shared host at isolation level L0. No ecbench sessions were used.

Each item's own `report.pdf`, `RESULTS.md` and `REDTEAM.md` hold its full data, intervals, deviations and reproduce commands.